Project - Molecular dynamics simulation: **Liquid water solvating a single graphene layer**

Objectives: Upload a water-graphene interface, run a full MD simulation and analyze microscopical structure and dynamics of water.

Tools: VMD (data visualization and analysis), NAMD (MD simulation)

STEP 1: Preparing Setup. Install and open VMD according the operating system of your computer —> http://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=VMD

NOTE: The first time you'll need to login as a new user into the page:
http://www.ks.uiuc.edu/Research/vmd/

1. Open VMD and upload your initial setup. The hydrophobic surface will be a single graphene layer of 20 x 20 Angstrom. It is solvated in KCl-water at 0.15 M concentration and at 298 K.

1. You should upload files 'charmm-gui-Water-graphene.tar' and unzip it.
2. Use the instructions: call 'vmd ';

   a. File -> New molecule -> Browse: Go to -> charmm-gui-Water-graphene/namd and select 'step3_input.psf' -> Load; and then do
   b. Browse: 'step3_input.pdb -> Load...and have a look.
   c. HINT1: Try to use the tools "Graphics -> Representations -> etc. Can use the VMD user guide: https://www.ks.uiuc.edu/Research/vmd/current/ug/
   d. HINT2: it is clearer if using "Display" -> Orthographic

STEP 2: Equilibration. Install and open NAMD according the operating system of your computer —> http://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=NAMD

Make sure it works by trying: > namd2 or ./namd2

1. Series of jobs:

   Go to -> charmm-gui-Water-graphene/namd

   Make "./namd2 step4_equilibration.inp"
2. This will run 125000 steps' energy minimization + MD, it may take 1 hour, depending on architecture and processor of your laptop. After this run, the equilibration should be ready.

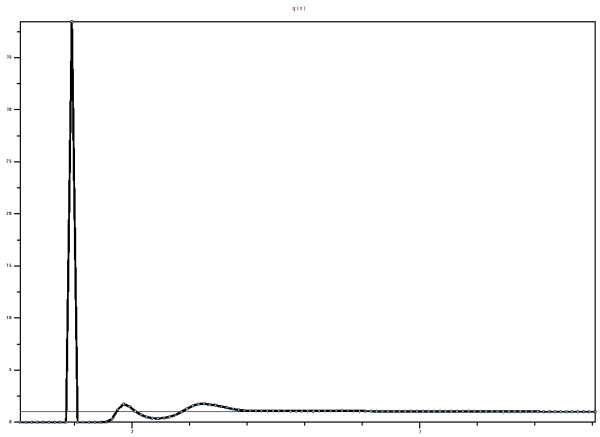
STEP 3: MD Production run and calculation of physical properties. After the system is equilibrated, a long run is in order. Use for instance 500000 steps. Get a (big) refreshment!

1. Now you should make the production run: "./namd2 step5_production.inp > output.log &" (500000 steps, total length 1 ns, if time step is set to 2 fs). Several hours. Estimate how long will it take before running it, trying a short test run.
2. After the long run ends, have a look at the final movie. To visualize it, use VMD: /home/…/charmm-gui/vmd/ step3_input.psf ./namd/step5_production.dcd &

Check the energy, temperature and pressure versus time. With the "log" file "output.log", you can use the tool "NAMD plot". You should get energy and temperature fluctuating around given mean values. What happens to pressure? Explain the results.

a. **Radial distribution functions of O-O, O-H, H-H, C-O, C-H:** Use 'Extensions' -> 'Analysis' -> 'Radial Pair Distribution Function g(r)'. In 'Selection 1' and '2', write: type 'OT' for g(r)_{OO}. The

numbers of frames run from 0 to N-1 (according the count shown at the VMD main screen as 'Frames'). For hydrogens use type 'HT' and type 'CA' for carbons. As an example, here is the rough g_{OH}(r):



b. **Coordination numbers**: The n(r) corresponding to the above g(r) may be obtained with the same tool ('Radial...') using 'Display Int(g(r))'.

Describe your results in physical terms.

c. **Hydrogen bonds in water:** Using the tool Extensions' -> 'Analysis' -> 'Hydrogen Bonds', calculate the number of HB per water molecule in the system. Set 'Selection 1' as 'resname TIP3' and set HB definition to 'Donor-Acceptor distance = 3.6 Å' and 'Angle cutoff = 30'.

Knowing that bulk water has about 4 HBs per molecule, explain your results.

d. **Water self-diffusion:** We can evaluate water diffusion by means of the 'RMSD Trajectory Tool'. We learned that:

$D = (1/6) \lim_{t \to \infty} RMSD(t)/t$

For the plot of RMSD(t), evaluate D.

FINAL REMARKS:

3. Play with 'Graphics' -> 'Representations' to customize your view.
4. You can save a snapshot of your system at 'File' -> 'Render'
5. And you can save a movie of your simulation with tool 'Movie Maker' from 'Extensions' -> 'Visualization'.

STEP 4: EXTENSIONS using CHARMM-GUI

1. In order to extend your study to other classes of systems, you must prepare a new input. The best tool for this is the facility “CHARMM-GUI” (https://www.charmm-gui.org). You must first create a user.
2. Go to ‘Input generator’ -> ‘Nanomaterial Modeler’ -> Select ‘Nanomaterial Type’ = ‘Carbonaceous Materials’ -> ‘graphene’. There you can select characteristics. Then take “solvated” and “next step”.
3. You can consider “Include ions” choosing “Fit waterbox size to material size” and ”next step” twice.
4. Select “Input generator” -> NAMD and go on.
5. Download “charm-gui.tar”, package including all files you need for the simulations.
6. First run file “step4_equilibration.inp” for equilibration.

7. Second, run file “step5_production.inp” for production.
8. Once finished, you’ll be able to analyze the new trajectory.

QUESTIONS:

1. Based on what you learned up to now, explore the **influence of temperature** on microscopic properties of water. Hint: Try equilibration + “short” production runs (about 300000 steps) at 283 and 323 K.

2. What is the effect of the **size** on your simulations? Try smaller cases (smaller or bigger graphene layer and less water). To do this, you should use the tool CHARMM-GUI.

3. How could you make the simulations more **realistic**? What particles could be included? Try a case (for instance N = 50, T = 303 K) including such particles. To do this, prepare a setup with CHARMM-GUI.